

Attestation-Gated Git Server

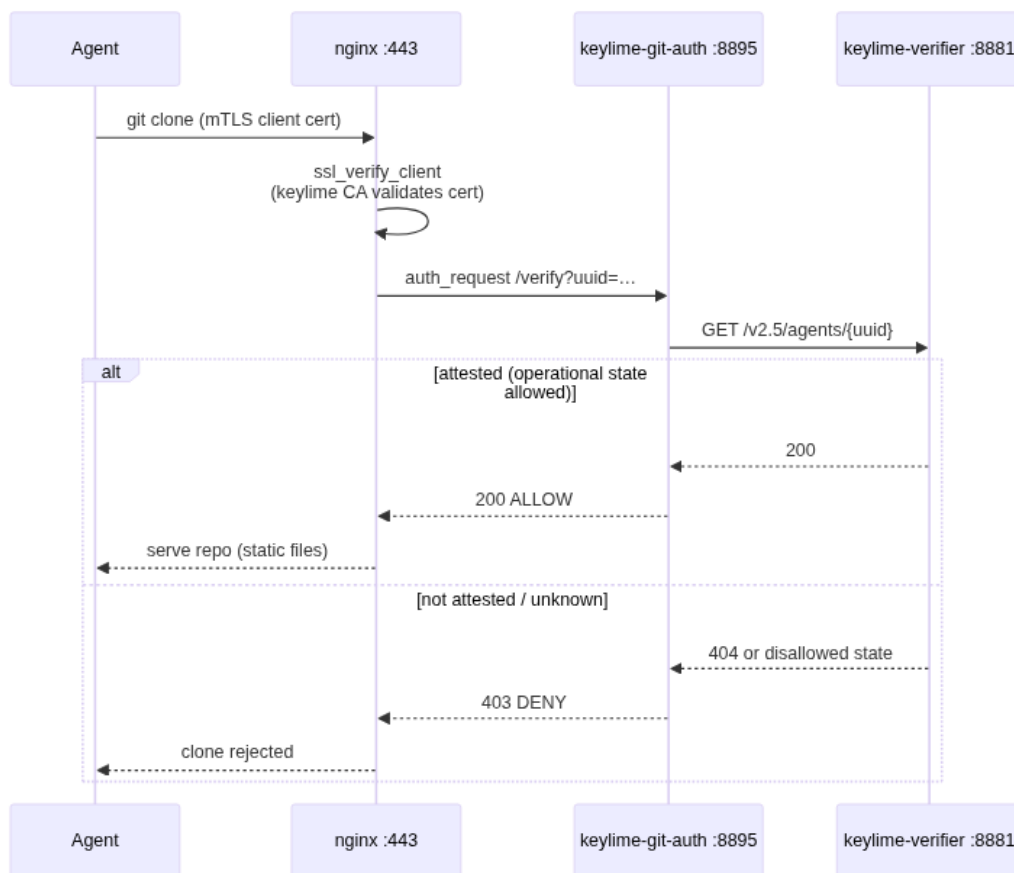
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Demo feature. This git server is designed for demonstration purposes: it shows that only TPM-attested builder machines can clone sensitive repositories. It uses git dumb HTTP (static file serving) which is adequate for small, infrequently-updated repos but not suitable for large-scale production use.

The git server lets attested builder machines clone repositories over HTTPS/mTLS. A machine that fails attestation, or has never been attested, cannot clone.

How It Works



Identity. nginx enforces mTLS against the keylime CA. The agent UUID (SHA-256 of the TPM Endorsement Key) is the CN of the client certificate, so no separate credential is needed: a machine provisioned into keylime automatically has its git identity.

Authorization. `keylime-git-auth` is a small Python daemon on `127.0.0.1:8895`. It queries the keylime verifier for the agent's operational state and only allows requests from agents in a known-good state (`START`, `SAVED`, `GET_QUOTE`, `GET_QUOTE_RETRY`, `PROVIDE_V`, `PROVIDE_V_RETRY`). Everything else is denied, including states it does not recognise. If the verifier is unreachable, it returns 503 and nginx rejects the clone (fail-closed).

Git transport. nginx serves the bare repo directory as static files (git dumb HTTP). No additional daemons are required. Run `git update-server-info` after each push, or install the provided post-receive hook, to keep the index files current.

Managing Repositories

Repositories are bare git repos in `/var/lib/keylime-git/repos/`.

Create a new repo:

```
git init --bare /var/lib/keylime-git/repos/my-repo.git
chown -R keylime:keylime /var/lib/keylime-git/repos/my-repo.git
git -C /var/lib/keylime-git/repos/my-repo.git update-server-info
ln -s /etc/keylime-git/hooks/post-receive \
    /var/lib/keylime-git/repos/my-repo.git/hooks/post-receive
```

Mirror an existing repo:

```
git clone --bare https://github.com/example/repo.git \
    /var/lib/keylime-git/repos/repo.git
chown -R keylime:keylime /var/lib/keylime-git/repos/repo.git
git -C /var/lib/keylime-git/repos/repo.git update-server-info
ln -s /etc/keylime-git/hooks/post-receive \
    /var/lib/keylime-git/repos/repo.git/hooks/post-receive
```

The post-receive hook runs `git update-server-info` automatically after every push so the index files stay current.

Cloning from an Agent

Builder machines include the `keylime-git-clone` wrapper, which automatically configures git with the machine's mTLS credentials:

```
keylime-git-clone https://<server-ip>/my-repo.git
```

This is equivalent to running `git clone` with the client certificate, key, and CA cert flags set. An optional destination directory can be appended:

```
keylime-git-clone https://<server-ip>/my-repo.git /path/to/dest
```

The credentials are provisioned automatically:

- **Client cert & key** are written to `/run/keylime-git/` by `report-measured-boot-state` after successful attestation.
- **CA cert** is written to `/run/keylime-git/ca-cert.pem` by `report-measured-boot-state` from `/boot/attestation-server.json`.

Both paths are world-readable, so any local user can run `keylime-git-clone` without elevated privileges.

Custom cert paths can be passed if needed:

```
keylime-git-clone --cert=/path/to/cert --key=/path/to/key --ca=/path/to/ca \
https://<server-ip>/my-repo.git
```

Verifying the Gate

Check both services are running:

```
systemctl status keylime-git-nginx keylime-git-auth
```

Confirm nginx rejects an unauthenticated request:

```
curl -k https://127.0.0.1:443/any-repo.git/info/refs
# Expected: 400 No required SSL certificate was sent
```

Watch the attestation gate in real time:

```
journalctl -fu keylime-git-auth
# Prints ALLOW / DENY with the agent UUID on every request
```

Nix Options

Option	Default	Description
<code>services.keylime.gitSerfalseenable</code>		Enable the git server
<code>services.keylime.gitSer443.port</code>		nginx HTTPS listen port
<code>services.keylime.gitSer8895authPort</code>		Auth daemon localhost port
<code>services.keylime.gitSer/var/lib/keylime-git/repos</code>		Bare repo root